Argo CD – Complete Detailed Notes in Telugu

1. Asalu Argo CD enduku use cheyali?
Avunu bro, real-time production lo Argo CD chala common ga use chestaru, especially **Kubernetes / EKS environments** lo.
First manam oka basic problem understand cheskovali. Suppose oka company daggara **50–100 microservices** unnayi. Prathi application ki deployment files, configurations, image versions, replicas, services ila chala Kubernetes resources untayi.
Every deployment ki manually:
kubectl apply
kubectl set image
kubectl rollout
kubectl check
cheyyadam possible, but applications ekkuva ayye koddi difficult and error-prone avtundi.
Evaraina wrong cluster ki command execute cheyyachu, wrong image deploy cheyyachu, or oka configuration change miss avvachu.
Argo CD tho Git lo deployment state maintain chesi, Kubernetes cluster ni automatic-ga aa state ki match chestaru.
Ante Git repository lo application ela undali ane **desired state** maintain chestam. Argo CD aa desired state ni Kubernetes cluster actual state tho continuously compare chestundi.

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2. Real-time Example 🔥
Suppose oka developer application code lo change chesadu.
Typical production flow ila undachu:
Developer
↓
GitHub
↓
CI Pipeline
(Jenkins / GitHub Actions)
↓
Docker Image
↓
Container Registry
↓
GitOps Repository
↓
Argo CD
↓
AWS EKS
↓
Pods
First developer code change GitHub ki push chestadu.
Tarvata Jenkins or GitHub Actions lanti CI pipeline code ni build chestundi, tests run chestundi, Docker image build chestundi.
Example ki Docker image:
myapp:v10
build ayindi.
Aa image Container Registry lo store avtundi.
Tarvata GitOps repository lo Kubernetes deployment configuration update chestaru:
image: myapp:v10
Ikkada GitOps repository lo unna configuration **Kubernetes lo application ela run avvali ane desired state** ni represent chestundi.
Argo CD aa Git repository ni observe chestundi and Git lo unna desired state prakaram Kubernetes/EKS cluster ni manage chestundi.

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3. Drift Example
Suppose Git lo:
image: myapp:v10
undhi.
But Kubernetes cluster lo accidentally:
myapp:v9
run avtundi.
Now situation:
Git:
myapp:v10

Kubernetes:
myapp:v9
Argo CD compare chesi difference ni identify chestundi.
Git = v10
Cluster = v9

→ Drift detected
Drift ante simple ga cheppalante:

Git lo manam expect chestunna state okati, Kubernetes cluster lo actual ga unna state inkokati.
Configured policy prakaram Argo CD aa difference ni sync/reconcile cheyyachu.
Automated sync/self-healing enabled unte cluster ni Git desired state ki automatically bring back cheyyagaladu.

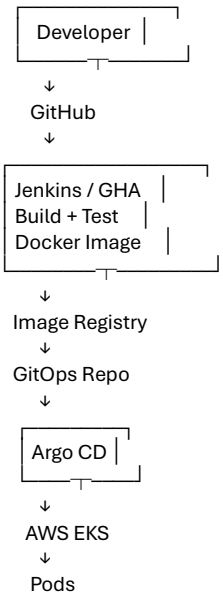
---

4. Mari Jenkins unte Argo CD enduku?
Idi **very important interview point**.
Jenkins and Argo CD rendu deployment ecosystem lo untayi, but rendu same work cheyyavu.
Jenkins responsibility
Jenkins ni mainly CI side lo think cheyyachu:
"Code build cheyyala? Test cheyyala? Docker image build cheyyala?"
Example:
Developer code
↓
Jenkins
↓
Build
↓
Test
↓
Docker Image
↓
Container Registry
Jenkins main focus **build, test, package** cheyyadam.
Argo CD responsibility
Argo CD ni Kubernetes CD/GitOps side lo think cheyyachu:
"Kubernetes lo Git lo unna desired version/state unda?"
Argo CD Git repository ni source of truth ga use chesi, Kubernetes cluster actual state ni desired state tho compare chestundi.
So rendu different responsibilities handle chestayi.
Simple ga gurthupettukovali ante:
Jenkins
→ CI
→ Build
→ Test
→ Docker Image

Argo CD
→ CD / GitOps
→ Kubernetes Deployment
→ Desired State
→ Actual State
→ Reconciliation

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5. Production lo Common Setup
Real production environment lo oka common architecture ila undachu:



Flow ni step-by-step ga chuste:

1. Developer code change chestadu.
2. Code GitHub ki push avtundi.
3. Jenkins / GitHub Actions build and tests run chestayi.

4. Docker image build avtundi.
5. Image Container Registry ki push avtundi.
6. GitOps repository lo required deployment configuration/version update avtundi.
7. Argo CD GitOps repository ni observe chestundi.
8. Argo CD desired state ni Kubernetes cluster ki apply/sync chestundi.
9. AWS EKS lo pods desired version tho run avtayi.

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6. One-line lo Argo CD Main Reason
Interview lo simple ga cheppali ante:
"Argo CD use cheyyadaniki main reason = Kubernetes deployments ni GitOps way lo automate, continuously reconcile, track and manage cheyyadam."
Ante deployment configuration Git lo centralized ga maintain cheyyachu.
Manual commands meeda dependency taggutundi, and cluster lo actual state Git desired state tho match avutundo continuously monitor cheyyachu.

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7. Real Companies Production lo Argo CD use chestara?
Yes — real companies production lo Argo CD use chestunnayi, particularly organizations running Kubernetes at scale.
Especially Kubernetes/EKS environments lo multiple applications and multiple environments manage cheyyalsi vachinappudu Argo CD useful avtundi.
But oka important point:
Every production environment lo Argo CD undadu.
Argo CD is one approach; organization architecture and requirements batti different tools or deployment methods use cheyyachu.
Examples:

- Jenkins / GitHub Actions direct deployment
- GitLab CI/CD
- Flux
- etc.

So interview lo **"Production lo Argo CD mandatory"** ani cheppakudadhu.
Correct answer:
"Argo CD is commonly used in production Kubernetes environments, especially at scale, but it is not mandatory. Other CI/CD and GitOps solutions are also used."

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8. DevOps Interview Perspective
DevOps interview lo Argo CD ni simple ga:
"Kubernetes CD/GitOps tool"
ga explain cheyyadam best.
Konchem detailed ga adigithe:
"Argo CD is a GitOps-based continuous delivery tool for Kubernetes. It uses Git as the desired-state source and continuously compares that desired state with the actual Kubernetes cluster state."
Inka follow-up vaste **drift detection, automated sync, self-healing, GitOps** concepts explain cheyyachu.

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9. Suppose YAML file rayachu kada? Mari Argo CD enduku?
Exactly bro — **YAML file tho kuda deployment cheyyachu.**
Anduke Argo CD first time confusing ga anipistundi.
But main difference:
"YAML unda leda?" kaadu.
Actual difference:
"YAML ni Kubernetes ki ela deploy/manage chestunnam?"
Kubernetes applications normally YAML manifests use chestayi.
Example:

```
spec:
  containers:
    - name: myapp
      image: myapp:v10
```

Ee YAML Kubernetes ki desired configuration cheptundi.
But aa YAML ni cluster ki **manual ga** kubectl **tho apply chestunnama**, leka **GitOps process dwara Argo CD manage chestunda** anedi actual difference.

---

10. Without Argo CD
Suppose manaki deployment.yaml undi:

```
spec:
  containers:
    - name: myapp
      image: myapp:v10
```

Manam manually:

```
kubectl apply -f deployment.yaml
```

chesam.
Appudu Kubernetes lo v10 deploy avtundi.
Initial ga everything correct ga undi.
But later evaro manually:

kubectl set image deployment/myapp myapp=myapp:v9
chesaru.
Now situation:
YAML file → v10
Kubernetes → v9
YAML file lo desired configuration inka v10 ga undi.
But cluster lo actual running configuration v9 ayipoyindi.

---

11. Kubernetes itself enduku automatically v10 ki tirigi velladu?
Idi important concept.
Kubernetes ki YAML file outside source lo undi ani automatic ga teliyadu.
Manam:
kubectl apply -f deployment.yaml
cheste Kubernetes aa configuration ni receive cheskoni cluster state maintain chestundi.
But later external ga YAML file change ayina, or someone manually cluster configuration ni modify chesina, Kubernetes Git repository ni continuously compare chesi:
"Git/YAML lo v10 undi, cluster lo v9 undi, kabatti v10 ki tirigi vellali."
ani automatically decide cheyyadu.
Kubernetes itself GitOps controller kaadu.
Anduke Argo CD lanti GitOps tool useful avtundi.

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12. Argo CD unte 🔥
Same YAML GitHub/GitOps repository lo untundi:

```
GitHub
└── deployment.yaml
    image: myapp:v10
```

Argo CD aa repository ni continuously monitor chestundi.
It checks:

```
Git desired state
   ↓
  v10

Cluster actual state
   ↓
  v9
```

Difference unte:
Drift detected
ani identify chestundi.
Argo CD ikkada Git lo unna state ni **desired state / source of truth** ga consider chestundi.

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13. Automated Sync / Self-Healing unte
If automated sync/self-healing enabled unte:

```
v9
↓
Argo CD detects drift
↓
Reconciliation
↓
v10 ki reconcile
```

So manual ga:
kubectl apply
cheyyalsina need chala varaku tagguthundi.
Important ga, Argo CD continuously desired state ni maintain cheyyadaniki try chestundi.

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14. Mari Kubernetes YAML alone saripoda?
Small Project
Yes bro.
Small project lo:
YAML + kubectl
perfectly enough.
Example ki few applications unte, oka developer/admin manual ga deployment manage cheyyadam easy ga untundi.
Appudu Argo CD add cheyyadam compulsory kaadu.

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15. Large Production Kubernetes Environment
Large environment lo situation different.
Suppose:

- 100s of YAMLs
- 20+ applications
- Multiple environments
- Multiple clusters

- Multiple teams
- Frequent deployments

Ila unnapudu manually:
kubectl apply -f ...
kubectl set image ...
kubectl rollout ...
manage cheyyadam difficult avtundi.
Different teams different clusters/environments lo deployments chestunte manual process inka complicated avtundi.
Ekkada deployment jarigindi, e version running lo undi, Git state and cluster state same ga unnaya, evaru change chesaru lanti questions kuda important avtayi.
Anduke Argo CD useful.

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16. One Super-Simple Comparison
YAML + kubectl
"Nenu cheppinappudu deploy cheyyi."
Ante manam command execute chesinappudu Kubernetes deployment perform chestundi.
YAML
↓
kubectl apply
↓
Kubernetes

Argo CD + Git
"Git lo unna state ni source of truth ga teesukoni, Kubernetes ni continuously aa state tho match cheyyi."
Ante Git lo desired state maintain chestam, Argo CD aa state ni continuously monitor/reconcile chestundi.
Git
↓
Argo CD
↓
Kubernetes
That's the real reason Argo CD use chestaru.

---

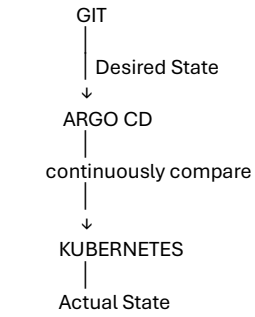
17. Very Important Correction – Drift Detection
Oka important point:
Argo CD lo **drift detection undadam alone** v9 ni automatically v10 ki marchadu.
Example:
Git = v10
Cluster = v9
↓
Argo CD
↓
Drift detected
Ikkada Argo CD difference ni detect chesindi.
But automatic ga cluster ni v10 ki change cheyyali ante appropriate configuration undali.
Automatic correction kosam:
Automated Sync / Self-Heal configuration
undali.

---

18. Automated Sync / Self-Healing Example
If automated sync/self-healing enabled:
Git
↓
v10
↓
Argo CD
↓
Cluster
↓
v10
Suppose someone manually cluster lo v9 ki change chesadu:
Git = v10
Cluster = v9
Argo CD:
Drift detected
↓
Automatic reconciliation
↓
Cluster → v10
So configured self-healing behavior valla manual change ni desired Git state ki bring back cheyyagaladu.

---

19. Final Understanding
Overall concept ni ila gurthupettuko:



Git lo:
myapp:v10
Cluster lo:
myapp:v10
unte:
Desired = Actual
Everything is in sync.
But:
Git = v10
Cluster = v9
unte:
Desired ≠ Actual
→ **Drift**
Argo CD drift ni identify chestundi.
Automated sync/self-healing configured unte:
Drift
↓
Reconcile
↓
Cluster → Desired State

---

20. Final Interview Summary
Argo CD is a Kubernetes CD/GitOps tool.
Kubernetes YAML lo **desired state** define chestam.
Without Argo CD:
YAML
↓
kubectl apply
↓
Kubernetes
Manam manually deployment trigger/manage cheyyali.
With Argo CD:
Git
↓
Desired State
↓
Argo CD
↓
Kubernetes
Argo CD Git ni **source of truth** ga teesukoni, cluster actual state ni desired state tho compare chestundi.
Difference unte **drift detect** chestundi.
Automated sync/self-healing enabled unte, cluster ni desired state ki automatically reconcile chestundi.
Therefore Argo CD main benefits:

- **GitOps** – Git ni desired-state source of truth ga use cheyyadam.
- **Automated Kubernetes deployments** – Manual deployment work tagguthundi.
- **Continuous reconciliation** – Git desired state and cluster actual state ni continuously compare chestundi.
- **Drift detection** – Git and cluster states different ga unte identify chestundi.
- **Self-healing** – Configuration enabled unte drift ni automatically correct cheyyagaladu.
- **Deployment tracking** – Applications/deployments state ni easier ga track cheyyachu.
- **Easier management at scale** – Many applications and environments ni manage cheyyadam easier avtundi.
- **Multiple applications/environments management** – Large Kubernetes environments lo centralized GitOps approach provide chestundi.

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21. One-Line Interview Answer

“Argo CD is a Kubernetes GitOps-based continuous delivery tool that continuously compares the desired state stored in Git with the actual state in Kubernetes and reconciles the cluster to the desired state. It is especially useful for managing deployments at scale.”

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22. Shortest Way to Remember
YAML + kubectl
=
Manual deployment

Git + Argo CD
=
GitOps + Continuous Reconciliation
Or even simpler:
YAML tells Kubernetes WHAT should run.
kubectl manually applies it.
Argo CD continuously makes Kubernetes MATCH what Git says should run.
Ila ippudu **shortcut notes kaakunda concept + explanation + examples + interview points** anni kalisi unnayi.

1.

ChatGPT said:

Sure bro. Ippudu **nee original 1–28 questions/headings ni exact ga same order lo** petti, **emi remove cheyyakunda**, prati topic ki konchem depth/explanation add chestunna. Style kuda nee sample lagaane untundi — Telugu + English technical terms, examples, diagrams, interview points.

Argo CD – Production & Interview Notes – Part 2

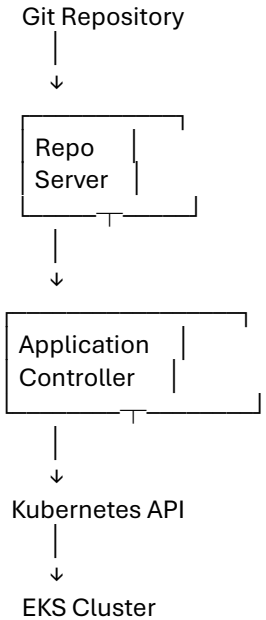
Inka konni important Argo CD concepts unnayi.

Argo CD – Production & Interview Notes – Part 2

1. Argo CD Architecture

Argo CD lopala multiple components untayi. Production lo Argo CD ela work chestundo understand cheyyali ante basic architecture telusukovadam important.

High-level architecture:



Ikkada simple ga flow enti ante, Argo CD Git repository nunchi application configuration ni retrieve chestundi. Aa desired configuration ni Kubernetes cluster lo unna actual state tho compare chestundi. Difference unte application OutOfSync ani identify cheyyachu. Configuration prakaram sync/reconciliation jarigite cluster desired state ki move avtundi.

Important Components

Argo CD API Server

API Server Argo CD ki main entry point laga work chestundi.

UI, CLI and API requests API Server dwara Argo CD ki vastayi.

Example:

Developer / DevOps

↓

Argo CD UI / CLI
↓
API Server
User application status chudali, sync cheyyali, application create cheyyali ante API Server handle chestundi.
For example, DevOps engineer Argo CD UI open chesi oka application OutOfSync lo undani chusi Sync click chesthe, aa request API Server dwara Argo CD components ki communicate avtundi.
So simple ga:
UI / CLI / API
↓
API Server
↓
Argo CD operations

Repository Server
Repository Server Git repositories nunchi application manifests ni fetch and process cheyyadaniki responsible.
Example:
Git Repository
↓
Repo Server
↓
Kubernetes YAML / Helm / Kustomize
Git lo YAML, Helm chart or Kustomize configuration unte, Repo Server aa configuration ni retrieve chesi Argo CD ki desired manifests provide chestundi.
Example:
Git
└── payment/
  ├── deployment.yaml
  ├── service.yaml
  └── configmap.yaml
Repo Server aa repository/path ni access chesi application ki required configuration ni Argo CD process ki available chestundi.
So Repo Server ni simple ga:
“Git nunchi application configuration/manifests ni retrieve and generate cheyyadaniki responsible component.”
ani gurthupettukovachu.

Application Controller
Idi Argo CD lo very important component.
Application Controller actual Kubernetes state ni Git desired state tho compare chestundi.
Example:
Git Desired State
↓
Application Controller
↓
Kubernetes Actual State
Suppose:
Git = v10
Cluster = v9
Controller difference detect chesi application OutOfSync ani identify chestundi.
Automated sync enabled unte required reconciliation kuda trigger chestundi.
For example:
Git = v10
Cluster = v9
↓
Application Controller

↓
Difference detected
↓
OutOfSync
Self-healing configuration kuda enabled unte, cluster lo manual change jarigina situation lo desired state ki return cheyyadaniki reconciliation jaragachu.
So Application Controller ni Argo CD lo **desired state vs actual state comparison and reconciliation logic ki central component** laga think cheyyachu.

Redis
Argo CD architecture lo Redis caching kosam use avtundi.
Frequently required information ni cache cheyyadam valla performance improve avvachu.
Simple ga interview lo:
“Redis is used by Argo CD for caching and improving performance.”
ani cheppachu.
Redis ni main deployment engine laga think cheyyakudadhu.
Simple ga:
Argo CD
↓
Redis
↓
Cache
Caching valla repeatedly same information retrieve/process cheyyalsina overhead konni situations lo reduce avvachu.

---

2. Argo CD Application
Argo CD lo Application ane concept chala important.
Application ante simply:
Git repository lo unna desired configuration ni particular Kubernetes cluster lo particular destination ki deploy/manage cheyyadaniki Argo CD ki ichina definition.
Application generally ee information ni specify chestundi:
Git Repository
↓
Repository Path
↓
Target Cluster
↓
Target Namespace
Example:
Git Repo:
github.com/company/gitops

Path:
apps/payment

Cluster:
EKS Production

Namespace:
payment
Argo CD Application ki ee information teliste, aa Git path lo unna application configuration ni specified cluster/namespace lo manage cheyyagaladu.
For example, company daggara:
payment
orders
users
ane applications unnayi anukundam.

Prathi application ki separate Argo CD Application definition create cheyyachu.
Argo CD
  │
  ├── payment-app
  ├── orders-app
  └── users-app
Prathi Application ki different Git path, namespace and destination cluster undachu.
So Application ane concept Argo CD lo **“what to deploy, where to get it from, and where to deploy/manage it”** ane information ni connect chestundi.

---

3. Argo CD Application Example
Conceptually Application configuration ila untundi:

```
apiVersion: argoproj.io/v1alpha1
kind: Application

metadata:
 name: payment-app

spec:
 source:
  repoURL: https://github.com/company/gitops
  path: apps/payment

 destination:
  server: https://kubernetes.default.svc
  namespace: payment
```

Ikkada:
repoURL
→ Git repository ekkada undi

path
→ Application manifests ekkada unnayi

destination
→ E Kubernetes cluster ki deploy cheyyali

namespace
→ E namespace lo deploy cheyyali
Real production lo Application resource ni use chesi Argo CD ki:
“Ee Git repository lo ee path lo unna configuration ni ee cluster/namespace lo manage cheyyi.”
ani cheptam.
Example:
Git Repository
└── apps/
  └── payment/
    ├── deployment.yaml
    ├── service.yaml
    └── configmap.yaml
Argo CD Application:
Source
→ apps/payment

Destination
→ Production EKS

Namespace

→ payment
Appudu Argo CD aa path lo unna desired configuration ni target cluster/namespace tho manage chestundi.
Production lo same repository lo multiple applications unte, prathi application ki different path use cheyyachu.

---

4. Sync ante enti?
Argo CD lo Sync very important concept.
Sync ante:
Git lo unna desired state ni Kubernetes cluster ki apply chesi states ni match cheyyadam.
Suppose:
Git:
image = v10

Cluster:
image = v9
Application:
OutOfSync
Now Sync chesthe:
Git v10
↓
Argo CD Sync
↓
Kubernetes v10
Appudu:
Desired State = Actual State
and application:
Synced
avvachu.
Example:
Git
↓
image: myapp:v10

Cluster
↓
image: myapp:v9

↓

Argo CD Sync

↓

Cluster
↓
image: myapp:v10
Sync ante simply “deployment” ani matrame anukokudadhu.
Argo CD context lo Sync means **desired configuration and cluster state ni reconcile/apply chesi match cheyyadam**.
Application lo multiple resources unte:
Deployment
Service
ConfigMap
Ingress
Sync operation valla required resources desired state ki move avvachu.

---

5. Manual Sync vs Automated Sync
Argo CD lo sync generally two ways lo jaragachu.
Manual Sync
Developer/DevOps person manually Sync click chestadu or CLI/API dwara sync trigger chestadu.
Flow:
Git
↓
Argo CD
↓
OutOfSync
↓
DevOps manually Sync
↓
Kubernetes
Idi controlled deployment environments lo useful avvachu.
For example production lo change approval complete ayyaka DevOps engineer manually Sync trigger cheyyachu.
Example:
Developer
↓
Git Change
↓
Argo CD detects
↓
OutOfSync
↓
Approval
↓
Manual Sync
↓
Production
Ila manual sync use chesthe deployment timing meeda human control ekkuva untundi.

Automated Sync
Automated Sync enabled unte Argo CD changes ni detect chesi automatically synchronization perform cheyyagaladu.
Flow:
Git change
↓
Argo CD detects
↓
OutOfSync
↓
Automatic Sync
↓
Kubernetes updated
So every time manually Sync click cheyyalsina need tagguthundi.
Example:
Git:
image = v10

Developer changes to:
image = v11

↓

Argo CD detects

↓

Automatic Sync

↓

Kubernetes:
image = v11
Production lo Auto Sync use chestunnappudu Git repository itself controlled and reviewed ga undadam important, because approved Git change automatic deployment ki lead avvachu.

---

6. Self-Healing ante enti?
Self-Healing Argo CD lo important production feature.
Suppose Git lo:
image = v10
and cluster lo kuda:
image = v10
Everything correct.
But someone manually cluster lo:
kubectl set image deployment/myapp myapp=myapp:v9
chesadu.
Now:
Git = v10
Cluster = v9
Argo CD drift detect chestundi.
If automated sync + self-healing configuration enabled unte:
Manual Change
↓
Cluster = v9
↓
Argo CD detects drift
↓
Self-Healing
↓
Cluster → v10
So self-healing purpose:
Manual or external changes valla cluster desired state nunchi deviate ayithe, configured behavior prakaram desired Git state ki tirigi reconcile cheyyadam.
Important point enti ante, self-healing **application crash ayithe magically fix cheyyadam** ani kaadu.
Main concept:
Desired State
↓
Drift
↓
Detect
↓
Reconcile
↓
Desired State
So Argo CD self-healing mainly **configuration drift correction** concept.

---

7. Automated Sync and Self-Heal same aa?
Same kaadu.
Rendu related concepts but exact same kaavu.
Automated Sync
Git lo desired state change ayithe cluster ni automatically sync cheyyadaniki help chestundi.

Example:
Git v10
↓
Git changed to v11
↓
Argo CD detects
↓
Automatic Sync
↓
Cluster v11
Ikkada source of change:
Git
Git desired state change ayyindi.

Self-Heal
Cluster lo manual/external change jarigi Git state tho mismatch ayithe automatically correct cheyyadaniki help chestundi.
Example:
Git v11
Cluster v11
↓
Someone manually changes cluster to v9
↓
Drift
↓
Self-Heal
↓
Cluster v11
Ikkada source of change:
Cluster
Cluster state manually/external ga change ayyindi.
Simple ga:
Auto Sync
= Git change → Cluster update

Self-Heal
= Cluster manual drift → Desired state ki return
Interview lo idi exact ga cheppagaligithe concept clear ani interviewer ki telustundi.

---

8. Prune ante enti?
Prune concept kuda important.
Suppose Git repository lo:
deployment.yaml
service.yaml
configmap.yaml
unnayi.
Argo CD avi cluster lo manage chestundi.
Later developer Git nunchi:
configmap.yaml
delete chesadu.
Now:
Git:
deployment.yaml
service.yaml

Cluster:
deployment

service
configmap
Cluster lo configmap inka undi.
Idi stale resource laga untundi.
Prune enabled unte, Git nunchi remove ayina resource ni cluster nunchi kuda remove cheyyagaladu.
Git resource deleted
↓
Argo CD detects
↓
Prune
↓
Cluster resource deleted
Prune ni carefully use cheyyali, because unintended Git deletion cluster resource deletion ki lead avvachu.
Example:
Developer mistake valla:
service.yaml
Git nunchi delete ayyindi.
Prune enabled unte:
Git deletion
↓
Argo CD
↓
Service deleted from cluster
Kabatti production lo Git repository changes controlled ga undali.
Simple ga:
Prune = Git desired state lo leni managed resources ni cluster nunchi remove cheyyadam.

---

9. Sync Status
Argo CD UI lo application ki different sync statuses kanipistayi.
Important statuses:
Synced
Git desired state and Kubernetes actual state match avutunnayi.
Git = v10
Cluster = v10

→ Synced
Ante configuration-wise cluster desired state tho match avtundi.

OutOfSync
Git and cluster states different ga unnayi.
Git = v10
Cluster = v9

→ OutOfSync
Idi drift or pending Git change indicate cheyyachu.
Example:
Git updated:
v10 → v11

Cluster:
v10
Argo CD:
OutOfSync
ani show cheyyachu.
Alage someone manually cluster ni modify chesina:
Git = v11

Cluster = v9
appudu kuda OutOfSync avvachu.

Unknown
Argo CD current synchronization state ni properly determine cheyyalekapothe Unknown status ravachu.
Usually repository access, cluster access or other communication-related issues investigate cheyyali.
Example:
Argo CD
↓
Cannot properly access/check
↓
Cluster / Repository
↓
Unknown
So Unknown ante application necessarily broken ani meaning kaadu. Argo CD ki current state ni properly determine cheyyadam possible kaaledu ani artham.

---

10. Health Status
Sync status and Health status same kaavu.
Sync status:
Desired configuration and actual configuration match avutunnaya?
Health status:
Application/resources actual ga healthy ga run avutunnaya?
Example:
Sync Status:
Synced

Health:
Degraded
Possible.
Ante configuration Git tho match avtundi, but application itself healthy ga ledu.
Important health states:

- Healthy
- Progressing
- Degraded
- Missing
- Unknown

Example:
Deployment desired replicas:
3 replicas
Git and cluster configuration match avvachu.
But pods:
1 Running
2 CrashLoopBackOff
unte application health degraded avvachu.
So:
Synced ≠ Always Healthy
Idi interview lo useful point.
Inka simple ga:
Sync Status
→ Configuration correct aa?

Health Status
→ Application actually healthy aa?
Example:
Git says:
3 replicas

Cluster configuration:
3 replicas

→ Synced

But:

Pod 1 → Running
Pod 2 → CrashLoopBackOff
Pod 3 → CrashLoopBackOff

→ Degraded
So Argo CD lo application ni evaluate chestappudu **Sync status + Health status rendu chudali**.

---

11. GitOps Repository Structure
Production lo Git repository ni proper structure tho maintain cheyyadam important.
Example:

```
gitops-repo/
│
├── apps/
│   ├── payment/
│   ├── orders/
│   └── user/
│
├── environments/
│   ├── dev/
│   ├── stage/
│   └── prod/
│
└── clusters/
    ├── dev-cluster/
    └── prod-cluster/
```

Actual structure organization batti change avtundi.
Main idea:
Application configurations ni Git lo organized ga maintain cheyyadam.
Multiple environments unte:
dev
stage
prod
different configurations maintain cheyyachu.
For example:
payment/
↓
dev
stage
prod
Dev lo:
replicas: 1
Production lo:
replicas: 5
lanti environment-specific differences undachu.
Proper repository structure valla:

- Applications easy ga identify cheyyachu
- Environment configurations separate ga manage cheyyachu
- Changes review cheyyachu

- Rollback easy ga trace cheyyachu

So GitOps repo just YAML storage kaadu; production lo adi **deployment configuration management system** laga important role play chestundi.

---

12. Helm with Argo CD

Production lo Argo CD only raw YAML tho matrame work cheyyadu.
Helm charts tho kuda Argo CD commonly use chestaru.
Flow:
Git
↓
Helm Chart
↓
Argo CD
↓
Rendered Kubernetes Manifests
↓
EKS
Suppose Helm values:

```
image:
  repository: myapp
  tag: v10
```

Argo CD Helm chart ni process chesi required Kubernetes manifests generate/deploy cheyyagaladu.
For example Helm chart lo:

```
templates/
  deployment.yaml
  service.yaml
```

and:
values.yaml
untayi.
Environment batti values change cheyyachu.
Example:
Dev:
image.tag = v10-dev

Prod:
image.tag = v10
Argo CD Helm chart ni render chesi Kubernetes ki final manifests provide chestundi.
So:
Argo CD can manage applications packaged using Helm.
Important ga remember cheyyali:
Helm
→ Application packaging / templating

Argo CD
→ GitOps deployment + reconciliation
Rendu responsibilities different.

---

13. Kustomize with Argo CD

Argo CD Kustomize ni kuda support chestundi.
Kustomize mainly base configuration ni reuse chesi environment-specific changes maintain cheyyadaniki useful.
Example:

```
base/
  deployment.yaml
  service.yaml

overlays/
```

dev/
stage/
prod/
Concept:
Base
↓
Dev Overlay → Dev configuration

Base
↓
Prod Overlay → Prod configuration
So same application configuration ni environments batti customize cheyyachu.
Example:
base:
replicas: 2
Production overlay lo:
replicas: 5
lanti change maintain cheyyachu.
Kustomize advantage enti ante same YAML ni copy-paste chesi multiple environment files create cheyyakunda, common base ni reuse chesi environment-specific patches/overlays maintain cheyyachu.
Argo CD aa Kustomize structure ni process chesi generated configuration ni Kubernetes cluster ki manage chestundi.

---

14. Helm vs Kustomize – Basic Understanding
Simple ga:
Helm
Templating + packaging approach.
Variables/values use chesi reusable charts create cheyyachu.
Example:
Chart
↓
values-dev.yaml
values-prod.yaml
Different values use chesi different environment configurations generate cheyyachu.
Kustomize
Existing Kubernetes YAML ni overlays/patches tho customize cheyyadam.
Example:
Base
↓
Dev Overlay

Base
↓
Prod Overlay
Same base configuration ni reuse chesi environment-specific changes maintain cheyyachu.
Both approaches ni Argo CD manage cheyyagaladu.
Interview lo:
“Argo CD supports multiple manifest generation approaches such as plain YAML, Helm and Kustomize.”
ani cheppachu.
Simple difference:
Helm
→ Template-based

Kustomize
→ Overlay / patch-based
But Argo CD ki rendu valid application source approaches.

---

15. Secrets – Important Production Concept

Kubernetes applications ki passwords, API keys, database credentials lanti secrets avasaram avvachu.
But plain-text secrets ni Git repository lo store cheyyadam security risk.
Example:
password: mypassword123
ila Git lo direct ga pettadam avoid cheyyali.
Production lo secrets kosam dedicated secret-management solutions use cheyyachu.
Common approaches include:
External Secret Management
↓
Kubernetes Secret
↓
Application
Environment and architecture batti tools/approaches change avtayi.
Important concept:
GitOps repo source of truth ayina, sensitive credentials ni plain text lo Git lo store cheyyakudadhu.
For example, database password Git repository lo:
password: production123
ani store cheyyadam dangerous.
Instead external secret-management system use chesi application ki required time lo secret provide cheyyachu.
Production architecture lo commonly secret lifecycle, access permissions, rotation and auditing kuda consider cheyyali.
So interview lo main point:
“We should not store plaintext credentials in Git. We should use an appropriate secret-management approach and integrate it with Kubernetes/Argo CD.”

---

16. RBAC in Argo CD

Production lo every developer ki complete Argo CD access ivvadam correct kaadu.
For example:
Developer
→ View application
→ Maybe sync dev

DevOps
→ Sync
→ Rollback
→ Manage applications

Admin
→ Full access
Argo CD lo RBAC use chesi users/teams ki permissions control cheyyachu.
Example concept:
Team A
→ Dev applications

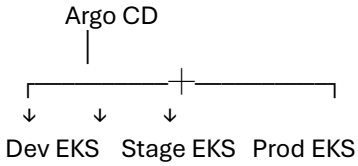
Team B
→ Production applications

Admin
→ All applications
Idi production security ki important.
Suppose developer ki production application delete permission unte accidental action valla major outage ravachu.
Kabatti least-privilege approach important.
Simple ga:

User
↓
Role / Policy
↓
Allowed Actions
↓
Allowed Applications / Projects
Permissions organization requirements batti define cheyyachu.
Interview lo:
“Argo CD RBAC is used to control who can view, sync, modify, or administer applications and projects.”
ani cheppachu.

---

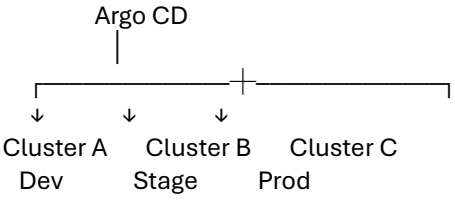
17. Multi-Cluster Deployment
Large organizations lo multiple Kubernetes clusters untayi.
Example:



Oka Argo CD setup multiple Kubernetes clusters ni manage cheyyagaladu.
Example:
Application A
→ Dev Cluster

Application B
→ Stage Cluster

Application C
→ Production Cluster
This is one of the reasons Argo CD large organizations lo useful.
For example company ki:
Dev EKS
Stage EKS
Production EKS
DR EKS
multiple clusters unte centralized Argo CD setup nunchi applications ni manage cheyyachu, architecture and access controls batti.
High-level:



Multi-cluster management valla every cluster ki separate manual deployment process maintain cheyyalsina burden reduce avvachu.

---

18. AppProject
Argo CD lo AppProject concept kuda important.
Application groups ki boundaries and permissions define cheyyadaniki AppProject use cheyyachu.
For example:
Project: Payments
↓
payment-dev
payment-stage

payment-prod
Project level lo:

- Allowed repositories
- Allowed clusters
- Allowed namespaces
- Resource restrictions
- Access policies

lanti controls define cheyyachu.
Simple ga:
Application = What application should Argo CD manage?
AppProject = That application can access/manage what and where?
Example:
Payments Project
↓
Allowed Git Repo
↓
payments-gitops
↓
Allowed Cluster
↓
Production EKS
↓
Allowed Namespace
↓
payment
AppProject concept especially multiple teams/applications unna organizations lo useful.
For example Payments team ki:
payment-dev
payment-stage
payment-prod
matrame access ivvachu.
Another team ki:
orders-*
applications matrame allow cheyyachu.
So AppProject ni **logical grouping + security boundaries + deployment restrictions** kosam think cheyyachu.

---

19. Rollback
Suppose production lo new version:
v10
deploy chesam.
But application lo serious issue vachindi.
Previous version:
v9
stable ga undi.
Deployment history maintain chesthe previous known-good version ki rollback cheyyadam possible.
Concept:
v8
↓
v9
↓
v10
↓
Issue
↓
Rollback
↓

v9
GitOps environment lo rollback process usually Git desired state ni appropriate previous version ki revert cheyyadam or Argo CD-supported history/sync mechanisms use cheyyadam based on the setup.
Important principle:
Rollback should also be controlled and traceable, rather than making undocumented manual cluster changes.
For example Git lo:
image: v10
change chesi:
image: v9
previous known-good state ki revert cheyyachu.
Then:
Git
↓
Argo CD
↓
Sync
↓
Kubernetes
↓
v9
Production lo rollback process already tested and documented ga undadam important.

---

20. Sync Waves and Hooks
Sometimes Kubernetes resources particular order lo deploy avvali.
Example:
Database migration
↓
Config
↓
Deployment
↓
Service
Random order lo resources apply chesthe application problem face cheyyachu.
Argo CD lo Sync Waves use chesi resource deployment order control cheyyachu.
Concept:
Wave 0
↓
Config / prerequisite

Wave 1
↓
Migration

Wave 2
↓
Application

Wave 3
↓
Other dependent resources
Suppose application deployment ki mundu configuration ready undali.
Appudu config ni earlier wave lo place cheyyachu.
Argo CD Hooks
Argo CD Hooks kuda support chestundi for specific lifecycle actions.
Example:
Before Sync

After Sync
Sync Fail
Use case batti appropriate hooks use cheyyachu.
Example:
Before Sync
↓
Database Migration
↓
Application Deployment
↓
After Sync
↓
Validation / Cleanup
So:
Sync Waves
→ Resource order control

Hooks
→ Specific lifecycle action execute cheyyadam
Complex deployment workflows lo rendu useful.

---

21. ApplicationSet
Suppose organization daggara:
50 clusters
100 applications
unnayi.
Prathi application kosam manually Argo CD Application resources create cheyyadam tedious.
ApplicationSet use chesi applications ni automatically generate/manage cheyyachu based on generators and patterns.
Concept:
Clusters
├── dev
├── stage
├── prod
└── another-cluster

ApplicationSet
↓
Generate Applications
↓
Argo CD Applications
Idi large-scale multi-cluster / multi-environment deployments lo very useful.
For example:
payment app
↓
payment-dev
payment-stage
payment-prod
Same pattern many applications and clusters ki apply cheyyachu.
Instead of manually:
Application 1
Application 2
Application 3
...
Application 100

create cheyyadam, ApplicationSet patterns/generators use chesi application resources generate cheyyachu.
So simple ga:
ApplicationSet = Many Argo CD Applications ni scalable and automated way lo generate/manage cheyyadaniki useful.

---

22. CI + Argo CD Complete Production Flow
Now entire concept ni connect cheddam.
Suppose developer application code change chesadu.
Developer
↓
GitHub
↓
Jenkins
↓
Build + Test
↓
Docker Image Build
↓
Container Registry
↓
GitOps Repo Update
↓
Argo CD
↓
EKS
↓
Pods
Detailed ga:
Step 1 – Developer
Developer code change chesi application repository ki push chestadu.
Example:
Developer
↓
git push
↓
GitHub
Step 2 – Jenkins
Jenkins webhook/trigger dwara pipeline start chestundi.
Build and tests run chestundi.
Build
↓
Unit Tests
↓
Integration Tests
Depending on pipeline, security scans and other quality checks kuda run avvachu.
Step 3 – Docker Image
Jenkins new Docker image create chestundi.
Example:
myapp:v11
Step 4 – Container Registry
Image registry ki push avtundi.
Registry
└── myapp:v11
Step 5 – GitOps Repository
Deployment configuration lo image version update avtundi:
image: myapp:v11

Idi GitOps repository lo commit/PR dwara update cheyyachu.
Important point:
CI pipeline image ni build chestundi; GitOps repository deployment lo e image version use cheyyalo represent chestundi.

Step 6 – Argo CD

Argo CD Git repository lo change detect chestundi.
Previous:
v10
New desired state:
v11
So application:
OutOfSync
avvachu.

Step 7 – Sync

Automated sync enabled unte Argo CD automatic ga sync chestundi.
Otherwise DevOps manually sync cheyyachu.
OutOfSync
↓
Manual / Automatic Sync
↓
Kubernetes

Step 8 – EKS

Kubernetes deployment update avtundi.
New pods:
myapp:v11
run avtayi.
So complete process:
Code Change
↓
CI
↓
Build/Test
↓
Docker Image
↓
Registry
↓
GitOps Repo
↓
Argo CD
↓
Kubernetes / EKS
Idi production lo **CI and CD responsibilities separate ga maintain chese common GitOps pattern**.

---

23. Argo CD Image Updater

One practical problem:
Developer new image build chestadu:
myapp:v11
But GitOps repository lo:
myapp:v10
undhi.
GitOps repo update cheyyali.
Ee image version update process ni automate cheyyadaniki Argo CD Image Updater lanti tooling use cheyyachu.
Concept:
New Image
↓

Container Registry
↓
Image Updater
↓
Git / Application configuration
↓
Argo CD
↓
Kubernetes
But actual architecture organization batti change avtundi.
Important point:
Argo CD itself primarily desired state reconciliation chestundi; image discovery/version update is a separate concern that can be automated using additional tooling.
For example:
Registry:
myapp:v10
myapp:v11
myapp:v12
New version available ayinappudu Image Updater configured strategy prakaram required image version ni identify chesi deployment configuration update process ni automate cheyyagaladu.
Then Argo CD normal GitOps flow prakaram new desired state ni reconcile chestundi.
So responsibilities:
CI
→ Build Image

Registry
→ Store Image

Image Updater
→ Image version update automation

Argo CD
→ Desired state reconciliation

---

24. Argo CD vs Jenkins
Interview lo frequently ask chestaru:
“Jenkins unte Argo CD enduku?”
Jenkins:
Mainly:
CI
↓
Build
↓
Test
↓
Package
↓
Docker Image
Argo CD:
Mainly:
CD / GitOps
↓
Git Desired State
↓
Kubernetes
↓
Continuous Reconciliation

Simple comparison:

Jenkins	Argo CD
CI-focused	Kubernetes CD/GitOps-focused
Build	Deploy/reconcile
Test	Desired vs actual state
Docker image creation	Kubernetes state management
Pipeline orchestration	GitOps reconciliation

Important:
Jenkins and Argo CD are not necessarily competitors. They can work together.
Real production flow:
Developer
↓
Jenkins
↓
Build + Test
↓
Docker Image
↓
Registry
↓
GitOps Repo
↓
Argo CD
↓
Kubernetes
Jenkins question:
“Can I build and test this application and create the artifact/image?”
Argo CD question:
“Does Kubernetes match the desired state defined in Git?”
So rendu different responsibilities handle chestayi.
One line lo gurthupettuko:
Jenkins
→ Build the application

Argo CD
→ Reconcile the Kubernetes deployment

---

25. Argo CD vs Flux
Both are popular Kubernetes GitOps approaches.
Argo CD
Strong points:

- Powerful UI
- Application-centric model
- Multi-cluster management
- Strong visibility
- Rich application management features

Flux
Flux is also a Kubernetes-native GitOps solution and is widely used.
High-level:
Git
↓
Flux
↓
Kubernetes
or:

Git
↓
Argo CD
↓
Kubernetes
Both tools main objective:
Git
↓
Desired State
↓
Kubernetes
Organization requirements batti choice change avtundi.
For example company ki:

- UI visibility important aa?
- Existing GitOps architecture enti?
- Team already Argo CD/Flux lo experienced aa?
- Multi-cluster requirements enti?
- Security and integration requirements enti?

ane factors consider cheyyachu.
Interview lo:
"Both Argo CD and Flux are GitOps tools for Kubernetes. Argo CD is particularly known for its application-centric model and UI."
ani cheppachu.
Main point:
Argo CD
→ GitOps + Application-centric management

Flux
→ GitOps + Kubernetes-native reconciliation
Exact choice organization architecture batti untundi.

---

26. Argo CD Production Challenges
Argo CD powerful tool ayina, production lo konni things carefully manage cheyyali.
Secrets
Sensitive information ni Git lo plain text lo store cheyyakudadhu.
Passwords, API keys, tokens and database credentials secure secret-management solution tho handle cheyyali.
RBAC
Wrong permissions valla users production applications ni accidentally modify cheyyakunda access control important.
Least privilege approach maintain cheyyali.
Git Repository Structure
Large number of applications/environments unte repository structure clean ga design cheyyali.
Otherwise configuration identify and maintain cheyyadam difficult avtundi.
Automated Sync
Production lo automated sync enable chesetappudu changes carefully review cheyyali.
Because Git change automatically production deployment ki lead avvachu.
Example:
Git change
↓
Auto Sync
↓
Production deployment
So production Git access and PR approval process important.
Prune
Prune enable chesinappudu Git resource deletion cluster resource deletion ki lead avvachu.
Kabatti carefully configure cheyyali.

Example:
Git resource deleted
↓
Prune
↓
Cluster resource deleted

Multi-Cluster

Multiple clusters manage chesthe cluster access, permissions and application boundaries properly maintain cheyyali.
Wrong cluster ki application deploy avvakunda destination configuration and access controls carefully manage cheyyali.

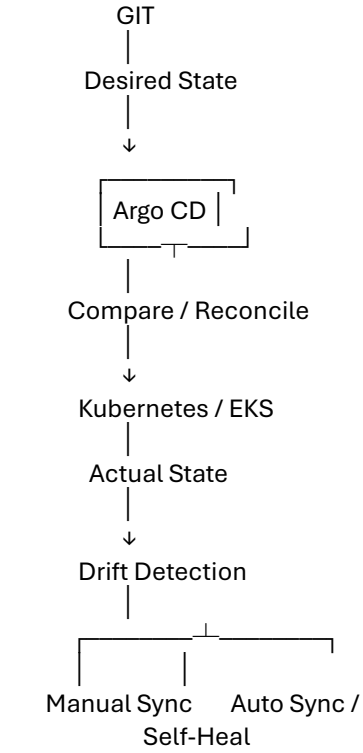
Rollback

Rollback process predefined and tested ga undali.
Production incident time lo rollback ela cheyyali ane clarity team ki mundhe undali.

---

27. Complete Argo CD Mental Model

Entire Argo CD ni one picture lo think cheyyali ante:



Git:

What should exist?

Argo CD:

Is the cluster matching Git?

Kubernetes:

What is actually running?

Complete mental model:
Git
↓
Desired State

Argo CD
↓
Compare

Kubernetes
↓

Actual State

Difference?
↓
OutOfSync / Drift

Configured reconciliation
↓
Cluster moves toward desired state
Example:
Git:
myapp:v10

Cluster:
myapp:v9

↓

Argo CD

↓

OutOfSync
Auto Sync/Self-Healing configuration applicable ayithe:
Argo CD
↓
Reconcile
↓
Cluster → v10
So Argo CD ni simply:
“Kubernetes deployment tool”
ani matrame anukokudadhu.
Better definition:
“Argo CD is a GitOps continuous delivery tool for Kubernetes that continuously compares desired state in Git with actual cluster state and reconciles them according to configuration.”

---

28. Complete Real-Time Example
Suppose production application:
myapp:v10
Current GitOps configuration:
image: myapp:v10
replicas: 3
Argo CD application status:
Synced
Healthy
Developer new code release chesadu.
CI:
Build
↓
Test
↓
Docker Image
↓
myapp:v11
GitOps repo update:
image: myapp:v11
replicas: 3

Argo CD detects:
Git:
v11

Cluster:
v10
Therefore:
OutOfSync
If automated sync enabled:
Argo CD
↓
Sync
↓
Kubernetes
↓
Rolling Update
↓
v11 Pods
After successful deployment:
Git:
v11

Cluster:
v11

Sync:
Synced

Health:
Healthy
Now suppose someone manually changes production to:
v9
Argo CD sees:
Git = v11
Cluster = v9
Drift detected.
If self-healing is enabled:
Drift
↓
Reconciliation
↓
Cluster → v11
So complete real-time production scenario:
Developer
↓
Code Change
↓
CI / Jenkins
↓
Build + Test
↓
Docker Image v11
↓
Container Registry
↓
GitOps Repo
↓

Argo CD
↓
Detect OutOfSync
↓
Sync
↓
EKS
↓
Rolling Update
↓
Pods v11
↓
Healthy + Synced
And later manual drift jarigithe:
Cluster manually changed
↓
Drift
↓
Argo CD detects
↓
Self-Healing
↓
Desired Git state
This is the core production behavior.

Part 2 – Interview Quick Revision
Last minute interview revision kosam entire Part 2 ni ila gurthupettuko:
Argo CD Architecture
→ API Server
→ Repo Server
→ Application Controller
→ Redis

Application
→ Source + Destination

Sync
→ Desired state → Cluster

Manual Sync
→ Human triggers sync

Auto Sync
→ Git change → Automatic sync

Self-Healing
→ Cluster drift → Desired state

Prune
→ Git lo deleted resource → Cluster lo remove

Sync Status
→ Synced / OutOfSync / Unknown

Health
→ Healthy / Progressing / Degraded / Missing / Unknown

GitOps
→ Git = Desired State

Helm
→ Templating + Packaging

Kustomize
→ Base + Overlays / Patches

Secrets
→ Never store plaintext sensitive data in Git

RBAC
→ Control user/team permissions

Multi-Cluster
→ One Argo CD → Multiple Kubernetes clusters

AppProject
→ Application boundaries + permissions

Rollback
→ Previous known-good version/state

Sync Waves
→ Resource deployment order

Hooks
→ Lifecycle actions

ApplicationSet
→ Generate/manage many Applications

Jenkins
→ CI / Build / Test / Image

Argo CD
→ Kubernetes CD / GitOps / Reconciliation

Flux
→ Another Kubernetes GitOps solution

Image Updater
→ Automate image version update

Core Concept
→ Git Desired State
→ Argo CD Compare
→ Kubernetes Actual State
→ Drift Detection
→ Reconciliation
Final ga one sentence:
Argo CD production lo main ga Kubernetes deployments ni GitOps way lo manage cheyyadaniki use chestam. Git desired state ni source of truth ga teesukoni, Kubernetes actual state tho continuously compare chesi, OutOfSync/drift detect chesi, configured sync/self-healing behavior prakaram cluster ni desired state ki reconcile chestundi.